

112 Autumn

EEIE30076

系統晶片設計 SOC Design

Lab. #2

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系級：機械所碩一

一、 **Brief introduction about the overall system**

學習兩種端口 (AXI-Master/AXI-Stream) 將資料從 PS (Processing System) 傳送至 PL (Programable Logic)。

二、 **What is observed & learned**

- **Differences between MAXI and Stream interface**

AXI-Master:

傳輸須帶有地址，可傳至特定的地址

可傳送指標/陣列參數

輸入/輸出地址必須要是連續記憶體地址

一條端口僅能一個讀+寫在一次的迴圈裡

AXI-Stream:

傳輸不帶有地址

不能傳送指標陣列參數

可無限制突發傳送 (burst transfer)

- **Differences between csim and cosim**

C Simulation:

僅驗證 (Validation) Code 是否可以通過 testbench 的測試集

C Co-simulation:

確認 (Verification) Code 生成出來的電路是否符合延遲時間

三、Screen dump

● Performance

```
1 =====
2 == Vitis HLS Report for 'fir_n11_maxi'
3 =====
4 * Date:      Tue Sep 26 20:08:58 2023
5
6 * Version:    2022.1 (Build 3526262 on Mon Apr 18 15:48:16 MDT 2022)
7 * Project:    Lab2_FIRN11MAXI
8 * Solution:   solution1 (Vivado IP Flow Target)
9 * Product family: zynq
10 * Target device: xc7z020-clg400-1
11
12
13 =====
14 == Performance Estimates
15 =====
16 + Timing:
17   * Summary:
18   +-----+-----+-----+-----+
19   | Clock | Target | Estimated| Uncertainty|
20   +-----+-----+-----+-----+
21   | ap_clk | 10.00 ns| 7.300 ns| 2.70 ns|
22   +-----+-----+-----+-----+
23
24 + Latency:
25   * Summary:
26   +-----+-----+-----+-----+-----+-----+
27   | Latency (cycles) | Latency (absolute) | Interval | Pipeline|
28   | min | max | min | max | min | max | Type |
29   +-----+-----+-----+-----+-----+-----+
30   | ?| ?| ?| ?| ?| ?| no|
31   +-----+-----+-----+-----+-----+-----+
32
33 + Detail:
34   * Instance:
35   +-----+-----+-----+-----+-----+-----+
36   | | | | | Latency (cycles) | Latency (absolute) | Interval | Pipeline|
37   | Instance | Module | min | max | min | max | min | max | Type |
38   +-----+-----+-----+-----+-----+-----+
39   | grp_fir_n11_maxi_Pipeline_XFER_LOOP_fu_242 | fir_n11_maxi_Pipeline_XFER_LOOP | ?| ?| ?| ?| ?| ?| no|
40   +-----+-----+-----+-----+-----+-----+
41
42   * Loop:
43   N/A
```

Performance (AXI-Master)

```
1 =====
2 == Performance Estimates
3 =====
4 + Timing:
5   * Summary:
6   +-----+-----+-----+-----+
7   | Clock | Target | Estimated| Uncertainty|
8   +-----+-----+-----+-----+
9   | ap_clk | 10.00 ns| 6.923 ns| 2.70 ns|
10  +-----+-----+-----+-----+
11
12 + Latency:
13   * Summary:
14   +-----+-----+-----+-----+-----+
15   | Latency (cycles) | Latency (absolute) | Interval | Pipeline|
16   | min | max | min | max | min | max | Type |
17   +-----+-----+-----+-----+-----+
18   | ? | ? | ? | ? | ? | ? | no|
19   +-----+-----+-----+-----+-----+
20
21 + Detail:
22   * Instance:
23   +-----+-----+-----+-----+-----+
24   | | | | | Latency (cycles) | Latency (absolute) | Interval | Pipeline|
25   | Instance | Module | min | max | min | max | min | max | Type |
26   +-----+-----+-----+-----+-----+
27   | grp_fir_n11_strm_Pipeline_XFER_LOOP_fu_112 | fir_n11_strm_Pipeline_XFER_LOOP | ? | ? | ? | ? | ? | ? | no|
28   +-----+-----+-----+-----+-----+
29
30   * Loop:
31   N/A
```

Performance (AXI-Stream)

● Utilization

```
1 -----
2 == Utilization Estimates
3 -----
4 * Summary:
5 -----
6 | Name | BRAM_18K | DSP | FF | LUT | URAM |
7 -----
8 | DSP | | - | - | - | - |
9 | Expression | | - | - | 0 | 40 | - |
10 | FIFO | | | | | |
11 | Instance | | 0 | 33 | 1886 | 2838 | - |
12 | Memory | | - | - | - | - | - |
13 | Multiplier | | - | - | - | 175 | - |
14 | Register | | - | - | 650 | - | - |
15 -----
16 | Total | | 0 | 33 | 4456 | 3053 | 0 |
17 -----
18 | Available | | 280 | 220 | 106480 | 53280 | 0 |
19 -----
20 | Utilization (%) | | 0 | 15 | 4 | 5 | 0 |
21 -----
22
23 * Detail:
24 * Instance:
25 -----
26 | Instance | Module | BRAM_18K | DSP | FF | LUT | URAM |
27 -----
28 | control_s_axi_U | control_s_axi | | 0 | 0 | 204 | 436 | 0 |
29 | grp_fir_all_maxi_pipeline_xfer_loop_fu_242 | fir_n11_maxi_pipeline_xfer_loop | | 0 | 33 | 2794 | 1884 | 0 |
30 | gmem_m_axi_U | gmem_m_axi | | 0 | 0 | 718 | 1318 | 0 |
31 -----
32 | Total | | | 0 | 33 | 1886 | 2838 | 0 |
33 -----
34
35 * DSP:
36 N/A
37
38 * Memory:
39 N/A
40
41 * FIFO:
42 N/A
43
44 * Expression:
45 -----
46 | Variable Name | Operation | DSP | FF | LUT | Bitwidth P0 | Bitwidth P1 |
47 -----
48 | lshl_int6_fu_289_p2 | | - | 0 | 0 | 48 | 33 | 2 |
49 -----
50 | Total | | | 0 | 0 | 48 | 33 | 2 |
51 -----
52
53 * Multiplier:
54 -----
55 | Name | LUT | Input Size | Bits | Total Bits |
56 -----
57 | an32Coef_address0 | 65 | 12 | 4 | 48 |
58 | ap_Ns_fsm | 65 | 15 | 1 | 15 |
59 | gmem_AVALID | 9 | 2 | 1 | 2 |
60 | gmem_BVALID | 9 | 2 | 1 | 2 |
61 | gmem_BREADY | 9 | 2 | 1 | 2 |
62 | gmem_BREADY | 9 | 2 | 1 | 2 |
63 | gmem_WVALID | 9 | 2 | 1 | 2 |
64 -----
65 | Total | | 575 | 17 | 10 | 73 |
66 -----
67
68 * Register:
69 -----
70 | Name | FF | LUT | Bits | Const Bits |
71 -----
72 | an32Coef_load_10_reg_446 | | 32 | 0 | 32 | 0 |
73 | an32Coef_load_1_reg_340 | | 32 | 0 | 32 | 0 |
74 | an32Coef_load_2_reg_350 | | 32 | 0 | 32 | 0 |
75 | an32Coef_load_3_reg_360 | | 32 | 0 | 32 | 0 |
76 | an32Coef_load_4_reg_370 | | 32 | 0 | 32 | 0 |
77 | an32Coef_load_5_reg_380 | | 32 | 0 | 32 | 0 |
78 | an32Coef_load_6_reg_390 | | 32 | 0 | 32 | 0 |
79 | an32Coef_load_7_reg_400 | | 32 | 0 | 32 | 0 |
80 | an32Coef_load_8_reg_410 | | 32 | 0 | 32 | 0 |
81 | an32Coef_load_9_reg_420 | | 32 | 0 | 32 | 0 |
82 | an32Coef_load_reg_330 | | 32 | 0 | 32 | 0 |
83 | ap_CS_fsm | | 14 | 0 | 14 | 0 |
84 | grp_fir_all_maxi_pipeline_xfer_loop_fu_242_ap_start_reg | | 1 | 0 | 1 | 0 |
85 | lshl_int6_cast_reg_440 | | 32 | 0 | 32 | 0 |
86 | ip32to64Input_read_reg_435 | | 64 | 0 | 64 | 0 |
87 | ip32to64Output_read_reg_430 | | 64 | 0 | 64 | 0 |
88 | trunc_int32_t_reg_451 | | 62 | 0 | 62 | 0 |
89 | trunc_int32_t_reg_455 | | 62 | 0 | 62 | 0 |
90 -----
91 | Total | | 658 | 0 | 658 | 0 |
92 -----
```

Utilization (AXI-Master)

```

1 =====
2 == Utilization Estimates
3 =====
4 * Summary:
5 +-----+
6 |      Name      | BRAM_18K| DSP | FF  | LUT | URAM|
7 +-----+
8 |DSP              |         | -|  -|  -|  -|  -|
9 |Expression       |         | -|  -|  0| 42|  -|
10 |FIFO             |         | -|  -|  -|  -|  -|
11 |Instance         |         | 0| 33| 2988| 1333| -|
12 |Memory           |         | -|  -|  -|  -|  -|
13 |Multiplexer      |         | -|  -|  -| 34|  -|
14 |Register         |         | -|  -| 36|  -|  -|
15 +-----+
16 |Total            |         | 0| 33| 3024| 1409| 0|
17 +-----+
18 |Available        |         | 280| 220| 106400| 53200| 0|
19 +-----+
20 |Utilization (%)  |         | 0| 15| 2| 2| 0|
21 +-----+
22
23 + Detail:
24 * Instance:
25 +-----+
26 |      Instance      |      Module      | BRAM_18K| DSP| FF  | LUT | URAM|
27 +-----+
28 |control_s_axi_U      |control_s_axi      |         | 0| 0| 154| 180| 0|
29 |grp_fir_n11_strm_Pipeline_XFER_LOOP_Fu_112 |fir_n11_strm_Pipeline_XFER_LOOP |         | 0| 33| 2834| 1153| 0|
30 +-----+
31 |Total                |                  |         | 0| 33| 2988| 1333| 0|
32 +-----+
33
34 * DSP:
35 N/A
36
37 * Memory:
38 N/A
39
40 * FIFO:
41 N/A
42
43 * Expression:
44 +-----+
45 |      Variable Name      | Operation| DSP| FF| LUT| Bitwidth P0| Bitwidth P1|
46 +-----+
47 |ret_v_fu_171_p2          |          | +| 0| 0| 40|          | 33|          | 2|
48 |grp_fir_n11_strm_Pipeline_XFER_LOOP_Fu_112_pstrnOutput_TREADY |          | and| 0| 0| 2|          | 1|          | 1|
49 +-----+
50 |Total                    |          |   | 0| 0| 42|          | 34|          | 3|
51 +-----+
52
53 * Multiplexer:
54 +-----+
55 |      Name      | LUT| Input Size| Bits| Total Bits|
56 +-----+
57 |ap_NS_fsm       | 25|          | 5| 1|          | 5|
58 |pstrnInput_TREADY_int_regslice | 9|          | 2| 1|          | 2|
59 +-----+
60 |Total           | 34|          | 7| 2|          | 7|
61 +-----+
62
63 * Register:
64 +-----+
65 |      Name      | FF | LUT| Bits| Const Bits|
66 +-----+
67 |ap_CS_fsm       | 4| 0| 4|          | 0|
68 |grp_fir_n11_strm_Pipeline_XFER_LOOP_Fu_112_ap_start_reg | 1| 0| 1|          | 0|
69 |tmp_reg_187     | 31| 0| 31|          | 0|
70 +-----+
71 |Total           | 36| 0| 36|          | 0|
72 +-----+

```

Utilization (AXI-Stream)

● Interface

```

1 =====
2 == Interface
3 =====
4 * Summary:
5 +-----+
6 |      RTL Ports      | Dir | Bits| Protocol | Source Object| C Type |
7 +-----+
8 |s_axi_control_AWVALID | in| 1|      s_axi| control| array|
9 |s_axi_control_AWREADY | out| 1|      s_axi| control| array|
10 |s_axi_control_AWADDR  | in| 7|      s_axi| control| array|
11 |s_axi_control_WVALID  | in| 1|      s_axi| control| array|
12 |s_axi_control_WREADY  | out| 1|      s_axi| control| array|
13 |s_axi_control_WDATA   | in| 32|     s_axi| control| array|
14 |s_axi_control_WSTRB   | in| 4|      s_axi| control| array|
15 |s_axi_control_ARVALID | in| 1|      s_axi| control| array|
16 |s_axi_control_ARREADY | out| 1|      s_axi| control| array|
17 |s_axi_control_ARADDR  | in| 7|      s_axi| control| array|
18 |s_axi_control_RVALID  | out| 1|      s_axi| control| array|
19 |s_axi_control_RREADY  | in| 1|      s_axi| control| array|
20 |s_axi_control_RDATA   | out| 32|     s_axi| control| array|
21 |s_axi_control_RRESP   | out| 2|      s_axi| control| array|
22 |s_axi_control_BVALID  | out| 1|      s_axi| control| array|
23 |s_axi_control_BREADY  | in| 1|      s_axi| control| array|
24 |s_axi_control_BRESP   | out| 2|      s_axi| control| array|
25 |ap_clk                | in| 1| ap_ctrl_hs| fir_nll_maxi| return value|
26 |ap_rst_n              | in| 1| ap_ctrl_hs| fir_nll_maxi| return value|
27 |interrupt              | out| 1| ap_ctrl_hs| fir_nll_maxi| return value|
28 |m_axi_gmem_AWVALID    | out| 1| m_axi| gmem| pointer|
29 |m_axi_gmem_AWREADY    | in| 1| m_axi| gmem| pointer|
30 |m_axi_gmem_AWADDR     | out| 64| m_axi| gmem| pointer|
31 |m_axi_gmem_AWID       | out| 1| m_axi| gmem| pointer|
32 |m_axi_gmem_AWLEN      | out| 8| m_axi| gmem| pointer|
33 |m_axi_gmem_AWSIZE     | out| 3| m_axi| gmem| pointer|
34 |m_axi_gmem_AWBURST    | out| 2| m_axi| gmem| pointer|
35 |m_axi_gmem_AWLOCK     | out| 2| m_axi| gmem| pointer|
36 |m_axi_gmem_AWCACHE    | out| 4| m_axi| gmem| pointer|
37 |m_axi_gmem_AWPROT     | out| 3| m_axi| gmem| pointer|
38 |m_axi_gmem_AWQOS      | out| 4| m_axi| gmem| pointer|
39 |m_axi_gmem_AWREGION   | out| 4| m_axi| gmem| pointer|
40 |m_axi_gmem_AWUSER     | out| 1| m_axi| gmem| pointer|
41 |m_axi_gmem_WVALID     | out| 1| m_axi| gmem| pointer|
42 |m_axi_gmem_WREADY     | in| 1| m_axi| gmem| pointer|
43 |m_axi_gmem_WDATA      | out| 32| m_axi| gmem| pointer|
44 |m_axi_gmem_WSTRB      | out| 4| m_axi| gmem| pointer|
45 |m_axi_gmem_WLAST      | out| 1| m_axi| gmem| pointer|
46 |m_axi_gmem_WID        | out| 1| m_axi| gmem| pointer|
47 |m_axi_gmem_WUSER      | out| 1| m_axi| gmem| pointer|
48 |m_axi_gmem_ARVALID    | out| 1| m_axi| gmem| pointer|
49 |m_axi_gmem_ARREADY    | in| 1| m_axi| gmem| pointer|
50 |m_axi_gmem_ARADDR     | out| 64| m_axi| gmem| pointer|
51 |m_axi_gmem_ARID       | out| 1| m_axi| gmem| pointer|
52 |m_axi_gmem_ARLEN      | out| 8| m_axi| gmem| pointer|
53 |m_axi_gmem_ARSIZE     | out| 3| m_axi| gmem| pointer|
54 |m_axi_gmem_ARBURST    | out| 2| m_axi| gmem| pointer|
55 |m_axi_gmem_ARLOCK     | out| 2| m_axi| gmem| pointer|
56 |m_axi_gmem_ARCACHE    | out| 4| m_axi| gmem| pointer|
57 |m_axi_gmem_ARPROT     | out| 3| m_axi| gmem| pointer|
58 |m_axi_gmem_ARQOS      | out| 4| m_axi| gmem| pointer|
59 |m_axi_gmem_ARREGION   | out| 4| m_axi| gmem| pointer|
60 |m_axi_gmem_ARUSER     | out| 1| m_axi| gmem| pointer|
61 |m_axi_gmem_RVALID     | in| 1| m_axi| gmem| pointer|
62 |m_axi_gmem_RREADY     | out| 1| m_axi| gmem| pointer|
63 |m_axi_gmem_RDATA      | in| 32| m_axi| gmem| pointer|
64 |m_axi_gmem_RLAST      | in| 1| m_axi| gmem| pointer|
65 |m_axi_gmem_RID        | in| 1| m_axi| gmem| pointer|
66 |m_axi_gmem_RUSER      | in| 1| m_axi| gmem| pointer|
67 |m_axi_gmem_RRESP      | in| 2| m_axi| gmem| pointer|
68 |m_axi_gmem_BVALID     | in| 1| m_axi| gmem| pointer|
69 |m_axi_gmem_BREADY     | out| 1| m_axi| gmem| pointer|
70 |m_axi_gmem_BRESP      | in| 2| m_axi| gmem| pointer|
71 |m_axi_gmem_BID        | in| 1| m_axi| gmem| pointer|
72 |m_axi_gmem_BUSER      | in| 1| m_axi| gmem| pointer|
73 +-----+

```

Interface (AXI-Master)



```
1 =====
2 == Interface
3 =====
4 * Summary:
5 +-----+-----+-----+-----+-----+-----+
6 |      RTL Ports      | Dir | Bits | Protocol | Source Object | C Type |
7 +-----+-----+-----+-----+-----+-----+
8 | s_axi_control_AWVALID | in | 1 | s_axi | control | array |
9 | s_axi_control_AWREADY | out | 1 | s_axi | control | array |
10 | s_axi_control_AWADDR | in | 7 | s_axi | control | array |
11 | s_axi_control_WVALID | in | 1 | s_axi | control | array |
12 | s_axi_control_WREADY | out | 1 | s_axi | control | array |
13 | s_axi_control_WDATA | in | 32 | s_axi | control | array |
14 | s_axi_control_WSTRB | in | 4 | s_axi | control | array |
15 | s_axi_control_ARVALID | in | 1 | s_axi | control | array |
16 | s_axi_control_ARREADY | out | 1 | s_axi | control | array |
17 | s_axi_control_ARADDR | in | 7 | s_axi | control | array |
18 | s_axi_control_RVALID | out | 1 | s_axi | control | array |
19 | s_axi_control_RREADY | in | 1 | s_axi | control | array |
20 | s_axi_control_RDATA | out | 32 | s_axi | control | array |
21 | s_axi_control_RRESP | out | 2 | s_axi | control | array |
22 | s_axi_control_BVALID | out | 1 | s_axi | control | array |
23 | s_axi_control_BREADY | in | 1 | s_axi | control | array |
24 | s_axi_control_BRESP | out | 2 | s_axi | control | array |
25 | ap_clk | in | 1 | ap_ctrl_hs | fir_n11_strm | return value |
26 | ap_rst_n | in | 1 | ap_ctrl_hs | fir_n11_strm | return value |
27 | interrupt | out | 1 | ap_ctrl_hs | fir_n11_strm | return value |
28 | pstrmInput_TDATA | in | 32 | axis | pstrmInput_V_data_V | pointer |
29 | pstrmInput_TVALID | in | 1 | axis | pstrmInput_V_dest_V | pointer |
30 | pstrmInput_TREADY | out | 1 | axis | pstrmInput_V_dest_V | pointer |
31 | pstrmInput_TDEST | in | 1 | axis | pstrmInput_V_dest_V | pointer |
32 | pstrmInput_TKEEP | in | 4 | axis | pstrmInput_V_keep_V | pointer |
33 | pstrmInput_TSTRB | in | 4 | axis | pstrmInput_V_strb_V | pointer |
34 | pstrmInput_TUSER | in | 1 | axis | pstrmInput_V_user_V | pointer |
35 | pstrmInput_TLAST | in | 1 | axis | pstrmInput_V_last_V | pointer |
36 | pstrmInput_TID | in | 1 | axis | pstrmInput_V_id_V | pointer |
37 | pstrmOutput_TDATA | out | 32 | axis | pstrmOutput_V_data_V | pointer |
38 | pstrmOutput_TVALID | out | 1 | axis | pstrmOutput_V_dest_V | pointer |
39 | pstrmOutput_TREADY | in | 1 | axis | pstrmOutput_V_dest_V | pointer |
40 | pstrmOutput_TDEST | out | 1 | axis | pstrmOutput_V_dest_V | pointer |
41 | pstrmOutput_TKEEP | out | 4 | axis | pstrmOutput_V_keep_V | pointer |
42 | pstrmOutput_TSTRB | out | 4 | axis | pstrmOutput_V_strb_V | pointer |
43 | pstrmOutput_TUSER | out | 1 | axis | pstrmOutput_V_user_V | pointer |
44 | pstrmOutput_TLAST | out | 1 | axis | pstrmOutput_V_last_V | pointer |
45 | pstrmOutput_TID | out | 1 | axis | pstrmOutput_V_id_V | pointer |
46 +-----+-----+-----+-----+-----+-----+

```

Interface (AXI-Stream)

● Co-simulation transcript/waveform

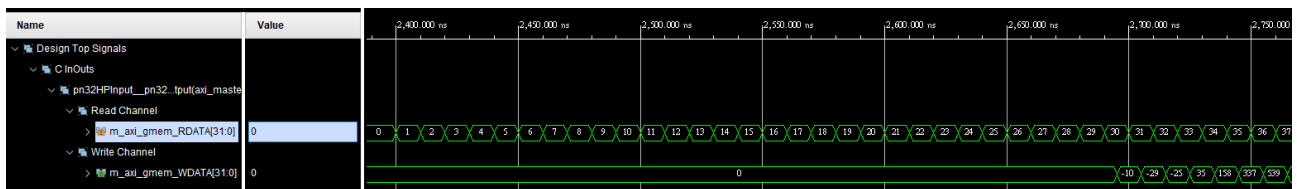
```

1 Report time      : Tue Sep 26 20:10:50 2023.
2 Solution         : solution1.
3 Simulation tool   : xsim.
4
5 +-----+-----+-----+-----+-----+-----+-----+-----+
6 |      |      |      |      |      |      |      |      |      |      |
7 +-----+-----+-----+-----+-----+-----+-----+-----+
8 |      |      |      |      |      |      |      |      |      |      |
9 +-----+-----+-----+-----+-----+-----+-----+-----+
10 |      |      |      |      |      |      |      |      |      |      |
11 |      |      |      |      |      |      |      |      |      |      |
12 +-----+-----+-----+-----+-----+-----+-----+-----+
13

```

			Latency(Clock Cycles)			Interval(Clock Cycles)			Total Execution Time
RTL	Status		min	avg	max	min	avg	max	(Clock Cycles)
VHDL	NA		NA	NA	NA	NA	NA	NA	NA
Verilog	Pass		943	943	943	NA	NA	NA	943

Co-simulation transcript (AXI-Master)



Co-simulation waveform (AXI-Master)

Index	Input	Output
0	1	0
1	2	-10
2	3	-29
3	4	-25
4	5	35
5	6	158
6	7	337
7	8	539

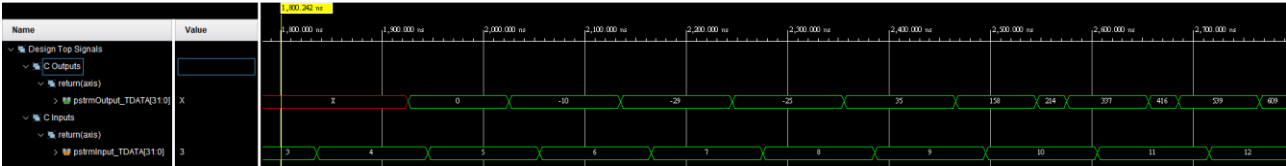
out_gold.dat

```

1 Report time      : Tue Sep 26 20:10:50 2023.
2 Solution         : solution1.
3 Simulation tool   : xsim.
4
5 +-----+-----+-----+-----+-----+-----+-----+-----+
6 |          |          |          |          |          |          |          |          |          |
7 + RTL + Status +-----+-----+-----+-----+-----+-----+ (Clock Cycles) +
8 |          |          | min | avg | max | min | avg | max |          |
9 +-----+-----+-----+-----+-----+-----+-----+-----+
10 | VHDL | NA | NA | NA | NA | NA | NA | NA | NA |
11 | Verilog | Pass | 943 | 943 | 943 | NA | NA | NA | 943 |
12 +-----+-----+-----+-----+-----+-----+-----+-----+
13

```

Co-simulation transcript (AXI-Stream)



Co-simulation waveform (AXI-Stream)

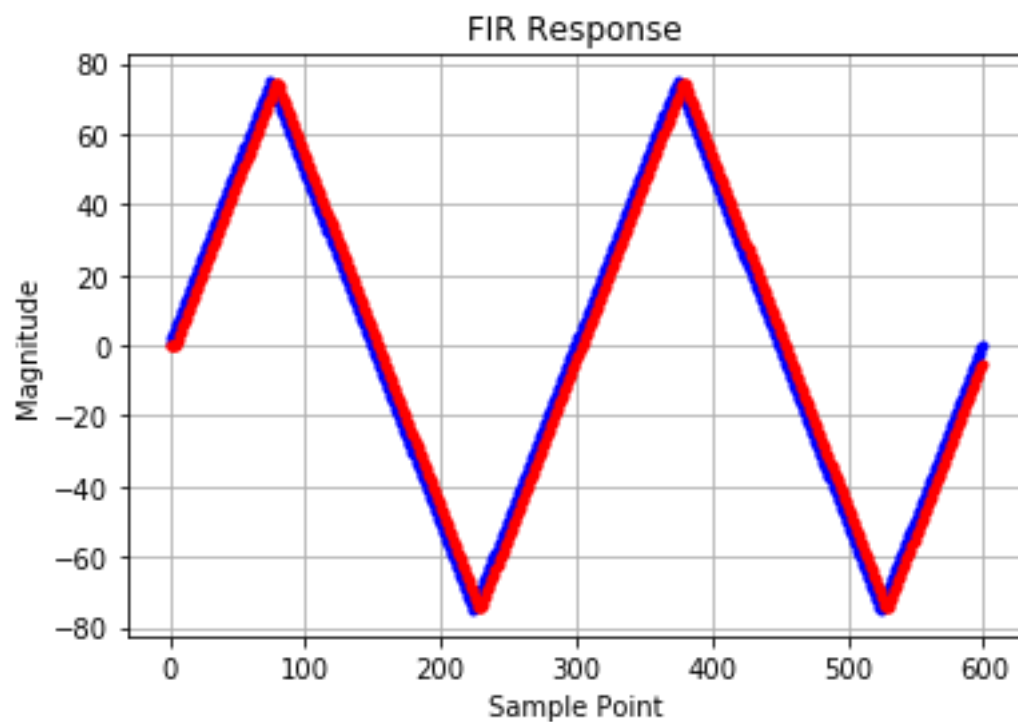
Index	Input	Output
0	1	0
1	2	-10
2	3	-29
3	4	-25
4	5	35
5	6	158
6	7	337
7	8	539

out_gold.dat

- Jupyter Notebook execution results
AXI-Master:



```
1 Entry: /usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py
2 System argument(s): 3
3 Start of "/usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py"
4 Kernel execution time: 0.0008635520935058594 s
```



AXI-Stream:



```
1 Entry: /usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py
2 System argument(s): 3
3 Start of "/usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py"
4 Kernel execution time: 0.001735687255859375 s
```

